# ASSIGNMENT – 02

**Q2. Analyze Oneplus Nord CE5 as an IoT edge device by identifying hardware, OS, and sensors. Hardware Operating System Sensors?**

The OneplusNord CE5 is a Reliable smartphone Which is a capable of acting as an IoT edge device which is responsible for collecting data from the surroundings, processing it locally, and interacting with the cloud or other devices.

Equipped with powerfull hardware with Android operating system, and several sensors, the Oneplus Nord CE5 contributes significantly to IoT operations, including smart home technology, health monitoring, and instant communication With IR Blaster Feature.

➢ **Hardware Components:**

The hardware enables sensing, processing, and communication.

- Processing Unit

CPU (Central Processing Unit) – Mediatek-Dimensity 8350 Apex, 3.35GHz

GPU (Graphics Processing Unit) – Arm Mali-G615 MC6

NPU/AI chip – Q1 Chip (For better Gaming Experience)

- Memory & Storage

RAM – 8GB

Internal Storage (Flash) – 128GB

➢ **Communication Modules:**

Cellular (4G/5G)

Wi-Fi 6

Bluetooth 5.4

NFC (for payments and device pairing)

GPS module

- **Power System:**

Battery + Power management IC (7001Mah)

- I/O Interfaces

Touchscreen display (Aqua Touch Display)

Camera modules(16 mp (Front), 50 MP (Back Camera))

Microphone & speakers (Single)

USB/charging port (Type-A to Type-C)

- **IoT Edge Role:**

Processes data locally ( face unlock, voice commands)

Communicates with cloud/IoT networks

- **Operating System (OS)**

Most smartphones run:

Android(16,17,18,19)

- **OS Responsibilities:**

Hardware abstraction (apps don't directly control hardware)

Resource management (CPU, memory, battery)

Security (permissions, sandboxing)

Networking (HTTP, Bluetooth, Wi-Fi communication)

- **IoT Edge Function:**

Runs edge applications (e.g., smart home apps, health monitoring)

Performs local data processing before sending to cloud

Enables real-time decision making

Sensors (Core of IoT Functionality)

Sensors allow smartphones to sense the environment, making them true IoT devices.

- ❖ Motion Sensors

Accelerometer – Detects movement, orientation

Gyroscope – Measures rotation

Magnetometer – Acts as a digital compass

- ❖ Environmental Sensors

Proximity sensor – Detects nearby objects (e.g., during calls)

Ambient light sensor – Adjusts screen brightness

Barometer (in some phones) – Measures air pressure

- ❖ Imaging & Audio

Camera – Image/video input

Microphone – Audio input

- ❖ Location & Connectivity

GPS sensor – Location tracking

- ❖ Biometric Sensors

Fingerprint sensor

Face recognition (camera-based or IR)

- ❖ IoT Edge Role:

Collect real-time data (motion, location, environment)

Enable smart applications (fitness tracking, navigatio**n,** automation

- ❖ **Final Analysis: (Why Oneplus Nord CE5= IoT Edge Device):**

A smartphone qualifies as an IoT edge device because it:

✓ Collects data via multiple sensors

✔ Processes data locally (AI, filtering, analytics)

✔ Communicates with cloud/IoT systems

✔ Supports real-time decision making

**Conclusion:**

The Oneplus Nord CE5 can serve as an effective IoT edge computing device owing to its advanced hardware components, optimized operating system, and numerous sensors.

- ✓ Hardware – Processing and Connectivity
- ✓ Operating System – Resource Management and Security
- ✓ Sensors – Data Collection

All of these features ensure that the device serves as a reliable IoT edge computing system.